Further paleomagnetic results from the Michikamau Intrusion, Labrador[1]

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Newly available isotopic age measurements and paleomagnetic results on rocks of the Michikamau Intrusion and its surroundings have made possible more precise estimates of the position and age of the paleomagnetic pole. Results from six new collecting localities, when combined with results from six different collecting localities reported earlier, provide a pole position at 1½° S, 142° W (A_{95} = 4½°). The age of crystallization of the intrusion is about −1460 Ma and the age of the pole is shown to be about −1450 Ma. The remanence is caused by fine grained, titanium-free magnetite and hematite.

De nouvelles déterminations d'âges isotopiques et des données paléomagnétiques sur des roches de l'intrusion de Michikamau et de ses environs rendent possible une évaluation plus précise de la position et de l'âge du pôle paléomagnétique. Les résultats de mesures à six nouvelles localités, lorsque combinées aux résultats de six localités différentes rapportées auparavant localisent le pôle à 1½° S, 142° O (A_{95} = 4½°). L'âge de cristallisation de l'intrusion est d'environ −1460 Ma et l'âge du pôle serait de −1450 Ma. La rémanence est causée par la présence de magnétite sans titane et d'hématite finement granulaires.

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Introduction

Magnetization directions have been measured in many Precambrian rocks dated by isotopic methods. Because of the prevalence of polycyclic deformation and metamorphism in the Precambrian, however, there are still relatively few rock units for which the ages of magnetization (as distinct from the depositional age or the metamorphic age of the rock) is known with confidence. The study reported here provides improved estimates of the age and position of the paleomagnetic pole for the Michikamau Intrusion.

The locations of the Michikamau Intrusion and other related plutons in central Labrador are shown in Fig. 1. The Michikamau Intrusion consists dominantly of weakly to strongly layered, leucotroctolite and coarse grained anorthosite, with a small amount of late adamellite (Emslie 1965, 1970). Surrounding the intrusion is a thermal contact aureole up to several kilometres wide, which is especially well developed in pelitic and semi-pelitic rocks of the Petscapiskau Group along the southern and eastern margin. The intrusion has not been subjected to any significant regional deformation or metamorphism since consolidation.

Recent isotopic age determinations have defined the time of emplacement and cooling of the Michikamau Intrusion fairly closely. A K–Ar hornblende age from marginal gabbro yielded −1479 ± 101 Ma (GSC 73-164, Wanless *et al.* 1974). A U–Pb zircon age of −1460 Ma (with an error of a few Ma) has been determined by Krogh and Davis (1973) from a sample of adamellite, the youngest member of the intrusive suite. Biotite from anorthosite gave a K–Ar age of −1400 ± 50 Ma (GSC 63-164, Wanless *et al.* 1965). Together, these ages imply that the intrusion was emplaced about −1460 Ma, and final cooling to low temperatures could conceivably have been extended until about −1400 Ma. Temperatures above the argon retention limit for biotite (approximately 200 to 300 °C, as suggested by Moorbath (1967) and Hanson and Gast (1967)) have not existed in the intrusion since that time. Hornblende from country rock amphibolite at site 14 (Fig. 2) yielded a K–Ar age of 2048 ± 54 Ma (GSC 73-163, Wanless *et al.* 1974), indicating that thermal effects of the intrusion were not sufficient to reset hornblende at a distance of 5 km from the contact.

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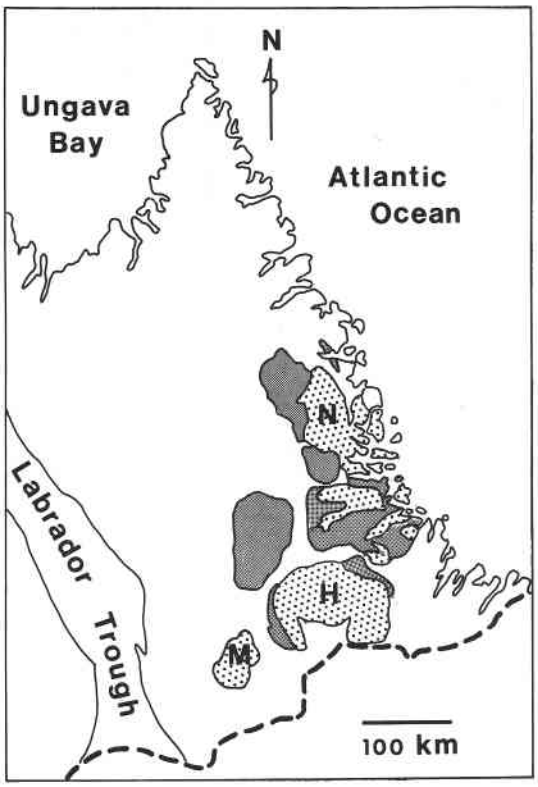


FIG. 1. Location of the Michikamau Intrusion and other related plutons in central Labrador. *M*—Michikamau Intrusion; *H*—Harp Lake Complex; *N*—Nain Anorthosite. Coarse stippled areas are anorthositic rocks, darker shading indicates associated adamellite group rocks. Heavy dashed line is the approximate position of the Grenville Front.

An earlier study by Murthy *et al.* (1968) showed that after cleaning in alternating fields[2] (AF) of 300 Oe, the rocks collected from six locations (Fig. 2) had a consistent WSW magnetization with low inclination. In a second paper, Murthy *et al.* (1970) have shown that remanent coercivities up to 1800 Oe are present, and they conclude that this very stable remanence is caused by fine grained needles of magnetite situated within the plagioclase. Our work generally confirms these conclusions except that, in addition to magnetite, hematite is also shown to be an important carrier of the stable remanence, and the hematite is sometimes reversed in direction from the magnetite. Our study of the country rocks also indicates that the remanence was probably acquired during cooling soon after emplacement, about −1460 Ma.

[2] 1 oersted $= \frac{1000}{4\pi}$ A/m

New Paleomagnetic Results

In 1971, one of us (R.F.E.) collected from five sites in the intrusion, seven sites in gneisses of the envelope within and near the thermal aureole around Fraser Lake, and one site in meta-andesite of the Petscapiskau Group 10 km to the south and outside the thermal aureole (Fig. 2). Very stable single component magnetizations predominate at the five sites sampled in the intrusion. Remanent coercive forces commonly exceed 1000 Oe, blocking temperatures are confined to the range 500 to 600 °C, and during demagnetization the directions undergo little change. A typical example is shown in Fig. 3. Over 60% of the remanence is retained at 600 °C, showing that hematite (presumably with a small proportion of ilmenite in solid solution) is the dominant carrier of the remanence. This was confirmed by a study of polished sections that show many large magnetite grains up to 1000 μm in diameter, with and without ilmenite lamellae; large ilmenite grains of comparable size, with fine elongate lamellae of hematite; and numerous small anisotropic grains that are probably hematite. The magnetite grains are too large to retain a stable remanence so that the latter two types of grains very probably carry most of the remanence. A further example is illustrated in Fig. 4, although at this site, additional more complex magnetizations are also present. One specimen underwent complete reversal during AF demagnetization. The westerly component, which comprises over 90% of the magnetization, has remanent coercivities of 750 Oe or less and is therefore probably due to magnetite. The easterly magnetization has remanent coercivities ranging from about 1000 to over 2900 Oe, and is therefore probably carried by hematite. In a second specimen, the magnetization underwent no change during thermal demagnetization up to 600 °C, but the blocking temperatures were predominantly between 550 and 600 °C. This indicates that magnetite carries most of the remanence, but hematite is again a significant contributor, because the direction is unchanged by 600 °C although the remaining magnetization is small. In this second specimen the magnetite and hematite are magnetized in the same direction. Study of polished sections shows that the

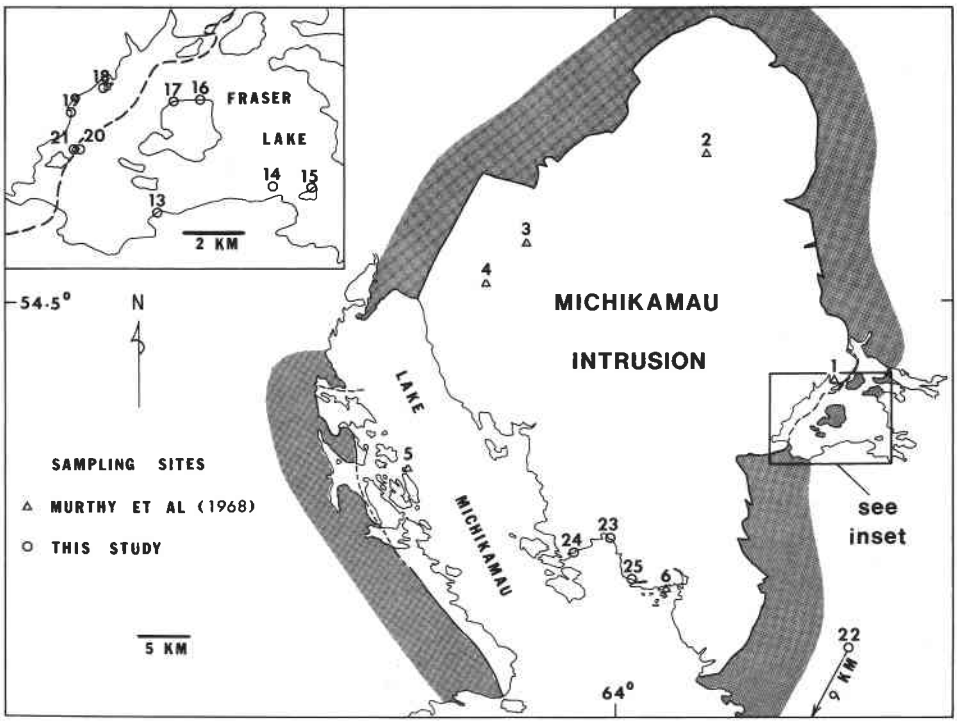


FIG. 2. Sampling localities in the Michikamau Intrusion. Detailed sampling around Fraser Lake is shown in the inset.

plagioclase is charged with minute opaque isotropic grains, which are presumably magnetite as described by Murthy *et al.* (1970), but some are anisotropic and are probably hematite. The anisotropic grains make up a few per cent of the total and this is consistent with the intensities observed in Fig. 4. Roquet (1954) measured saturation remanent magnetization values of 0.14 and 0.13 emu/g, respectively, for fine grained magnetite and hematite powders, so that relative intensities of about 0.05 are consistent with the presence of a few per cent of hematite. The absolute values given in Fig. 4 indicate the amount of hematite constitutes about 0.1% of the total rock; this is consistent with the optical observations. The examples shown in Figs. 3 and 4 are interesting cases of rocks in which magnetite is the most abundant magnetic mineral, but hematite contributes substantially to the remanence. A discussion of this phenomenon may be found in Irving (1964, p. 25).

The magnetization of the country rocks at Fraser Lake generally has low remanent coercive forces and blocking temperatures, with random directions. Only at site 13, 3 km from the contact, do coherent magnetizations occur (Figs. 2 and 5). The blocking temperatures are distributed between 100 and 400 °C, but remanent coercive forces up to 1250 Oe are present. The directions are the same as those in the intrusion. Large magnetite grains, large composite grains of hematite and ilmenite, and numerous patches of fine red dust, presumably hematite, are present in polished sections. The remanence properties are similar to those of viscous partial thermoremanence acquired by heating to temperatures below the Curie temperature for long periods.

The magnetization in country rocks at site 22, well outside the thermal aureole, is of some interest. A whole rock Rb–Sr isochron determined from low-grade meta-volcanic rocks at this locality yielded an age of -1559 ± 60 Ma ($\lambda = 1.39 \times 10^{-11}\,y^{-1}$; Emslie, *in* Wanless and Loveridge 1976). These rocks are considered to form part of the Petscapiskau Group, and are believed to be stratigraphically equivalent to part of the Croteau Group, the main outcrop area of which is 150 km to the east and which

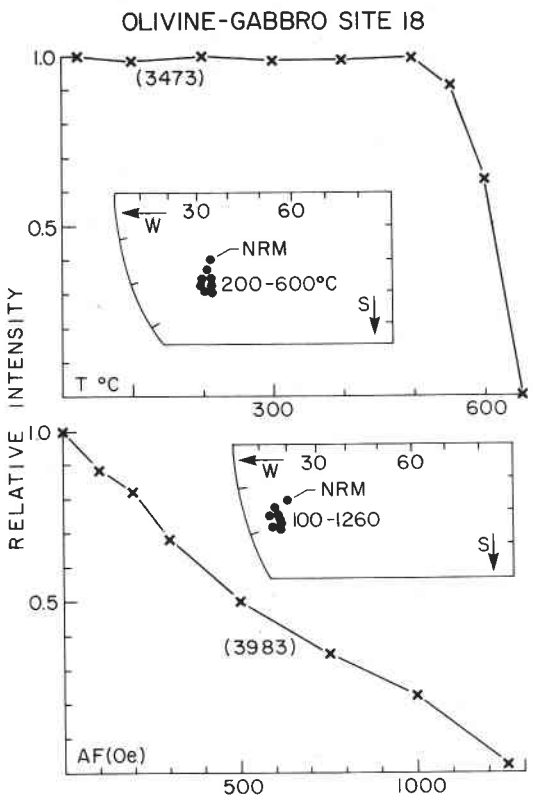


FIG. 3. Olivine gabbro, site 18. The decay of intensity during thermal and AF demagnetization and the corresponding directions. A single magnetization only is present. The initial (NRM) intensity in microgauss is given in brackets. Inclinations are downwards.

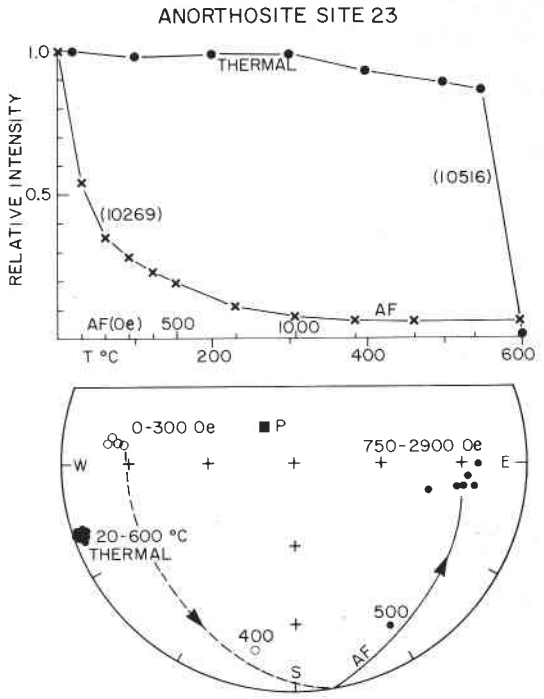


FIG. 4. Anorthosite, site 23. The decay of intensity during thermal and AF demagnetization and corresponding directions (below). A single magnetization only is present in the thermally demagnetized specimen. In the other specimen, two anti-parallel magnetizations are present with moderate and high remanent coercive forces, so that the specimen undergoes an exact reversal during AF demagnetization. The initial (NRM) intensity is given in microgauss. Dots (circles) indicate downward (upward) inclinations.

has yielded a nearly identical whole rock Rb–Sr isochron age of -1560 ± 42 Ma ($\lambda = 1.39 \times 10^{-11}$ y^{-1}; Wanless and Loveridge 1972).

The magnetization in rocks at site 22 is shown in Fig. 6. It has excellent end-points, for both AF and thermal demagnetization, with northeast directions. Remanent coercive forces up to 1250 Oe occur, and the blocking temperatures are between 550 and 600 °C. It is not certain whether the magnetization dates from the time of formation of the rocks or from the subsequent low-grade metamorphism, but for present purposes this is unimportant. The important point is that near the intrusion there are magnetization directions parallel to those in the intrusion, but outside the thermal aureole very different directions are found. The conditions for the classical contact test of Bruhnes (see Everitt and Clegg 1962) are satisfied, indicating that the westerly magnetization of the intrusion was probably acquired during initial

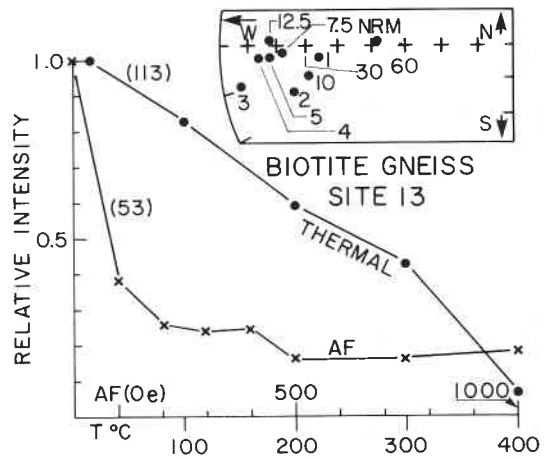


FIG. 5. Biotite gneiss, site 13. The decay of intensity during thermal and AF demagnetization. The corresponding direction changes during AF demagnetization are also shown. The initial (NRM) intensity is given in microgauss. All the inclinations are downwards.

TABLE 1. Remanent magnetization at each collecting locality

	Collecting site	UTM grid reference	N	M	$D°, I°$ (R)	S°	Treatment
18	olivine gabbro	E452800, N6029600	5	4459	249, +16(4.99)	04	1000 Oe, 550 °C
19	olivine gabbro	E451600, N6029000	5	3493	255, +19(4.96)	07	750 Oe, 600 °C
23	anorthosite	E433800, N6016400	3	12426	262, −01(2.92)	13	100 Oe, 550 °C
24	anorthosite	E430200, N6015000	3	12345	253, +03(2.86)	18	1000 Oe, 550 °C
25	anorthosite	E431300, N6022400	4	4916	261, +25(3.90)	13	300 Oe, 600 °C
13	biotite gneiss	E454700, N6025200	5	61	262, +13(4.75)	18	500 Oe
22	meta-andesite	E448500, N5992400	4	478	043, −28(3.97)	07	300 Oe, 500 °C

N is the number of separately oriented cores. M is the mean intensity of NRM in microgauss. $D°, I°$ is the mean direction, R the resultant, and S° the standard deviation. The cleaning procedure used is indicated in the last column. The pole for the meta-andesite (site 22) is 12° S, 106° W (9°, 5°).

TABLE 2. Summary of results

	B	N	$D_m°$,	$I_m°$	k	S°	$\alpha_{95}°$	Pole	$dm°, dp°$
Directions (Murthy *et al.* 1968)	6	29	262	+09½	35	13	10	½° S, 145½° W	10, 5
Directions, herein	6	25	257	+12½	53	10	9½	2½° S, 139½° W	9½, 5
Directions, total	12	54	259½	+11	43	12	6½	1½° S, 142½° W	7, 3½
Poles, total	12	54	—	—	103	8	4½	1½° S, 142° W	—

B is the number of collecting localities. N is the total number of oriented cores. $D_m°$, $I_m°$ are the mean directions. k is Fisher's estimate of precision (Fisher 1953). S° is the standard deviation. $\alpha_{95}°$ is Fisher's cone of confidence. dm and dp are the polar errors.
In the last entry, the pole is calculated as the mean of poles from each locality and the error entered under α_{95}.

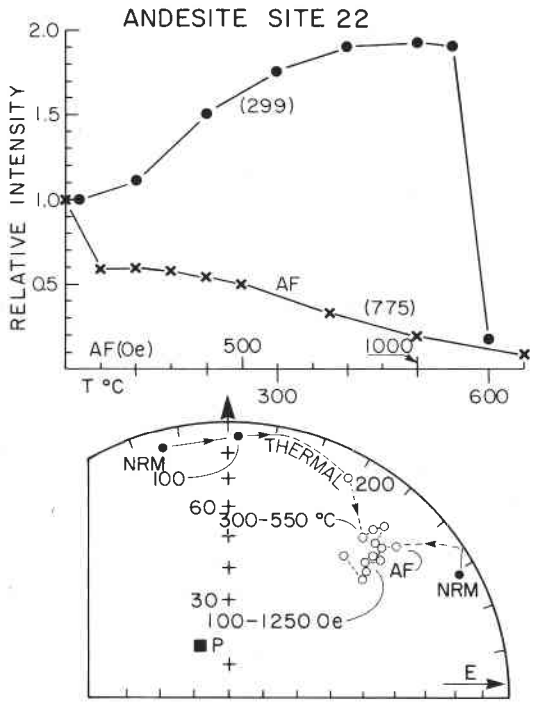


FIG. 6. Meta-andesite, site 22. The decay of intensity during thermal and AF demagnetization. The corresponding direction changes are shown below. The initial (NRM) intensity is given in microgauss. Dots (circles) indicate downward (upward) inclinations.

cooling. The presence of high discrete blocking temperatures at all sites studied in the intrusion is a further reason for believing this to be so, because remanence of this type can resist the temperature conditions associated with middle greenschist facies metamorphism (Pullaiah *et al.* 1975), and the Michikamau Intrusion has not been subjected to a metamorphism of this degree.

Discussion of Results

In Table 1 the results from each collecting locality are summarized, and in Table 2 they are compared to and combined with the earlier results. The agreement between our study and that of Murthy *et al.* (1968) is excellent. The combined results yield a pole with an error that is a factor of two less than that given by Murthy *et al.* (1968).

The U–Pb zircon age of −1460 Ma for the youngest member of the Michikamau Intrusion is very close to the U–Pb zircon age of −1450 Ma (Krogh and Davis 1973) from similar rocks of the Harp Lake Complex to the east (Fig. 1). It seems probable that these ages are close to the time of final crystallization of these intrusions, and thus set an older limit to the age of the magnetization. If a body cools slowly, however, the magnetization may be much younger than

the date of crystallization. Even an intrusion as large as Michikamau, however, is unlikely to require more than a few Ma to cool from emplacement temperatures of perhaps 1150 °C to below about 300 °C, providing that the regional environment was not at elevated temperatures. The fact that a different magnetization has been shown to exist in the older Petscapiskau Group (site 22) implies that regional heating, if it existed, was not of sufficient intensity to destroy that magnetization. It is inferred, therefore, that regional temperatures must have been relatively low (below 300 °C) at the time of intrusion and that the magnetization was acquired not more than a few Ma after crystallization. The thermal demagnetization characteristics provide additional evidence in support of this conclusion. The blocking temperatures are high and confined to a narrow range just below the Curie temperature; theoretical and experimental studies (Pullaiah *et al.* 1975) have shown that such remanence is expected to be older than K–Ar mica ages (biotite from Mickikamau is -1400 ± 50 Ma). It is therefore concluded that the most probable age of magnetization of the Michikamau Intrusion is about -1450 Ma. The upper and lower limits of this estimate are -1460 and -1400 Ma, respectively.

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